MIL-S-62359(AT)
29 July 1983

MILITARY SPECIFICATION

SEMITRAILER, BOTTOM DUMP: 18 TO 22 CUBIC YARD CAPACITY, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tandem axle, single and dual hopper, bottom dump semitrailers with minimum struck capacities of 18 to 22 cubic yards. Semitrailers procured under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

1.2 Classification. Semitrailers shall be of the types and sizes shown in table I, as specified (see 6.2):

TABLE I. Semitrailer classification.

Type	Nomenclature	Size
I	Single hopper, single gate, spreader type	18, 20
II	Single hopper, single gate, windrow type	18, 20
III	Dual hopper, dual gate, spreader type	20, 22
IV	Dual hopper, dual gate, windrow type	20, 22

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Size. Size of the semitrailer shall be determined by the total minimum struck capacity of the hopper(s) as follows:

Size 18 - 18 cubic yards
Size 20 - 20 cubic yards
Size 22 - 22 cubic yards.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL
ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway).

MILITARY
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

STANDARDS
MILITARY
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223 - Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking, and Data Plate Standards.
MS75021 - Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the, contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code - Section VIII.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers-Trucks, Truck-Tractors, and Trailers (DoD adopted).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DoD adopted).
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer (DoD adopted).
J741	- Capacity Ratings - Scraper, Dumper Body and Trailer Body (DoD adopted).
J1292	- Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.

(SAE Standards and Recommended Practices are available from SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.l.l.3, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, which meet or exceed the requirements of this specification. All vehicle items shall be as represented in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. Vehicles and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Treatment, painting, identification marking and data plates. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223.

3.1.1.3 Lifting and tying down attachments. When specified (see 6.2), the semitrailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to MIL-STD-209, type III. Attachments shall be indicated by a transportation plate conforming to MIL-P-514, except tying down attachments may be identified by stenciling or other suitable marking on the semitrailer. Tying down marking shall clearly indicate that the attachments are intended for tying down of the semitrailer on the carrier when shipped.

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3.2 Dimensions, clearances, payloads and capacities.

3.2.1 Dimensions and clearances. The semitrailer, without payload, uncoupled from the towing vehicle, and resting on its landing less on level ground, shall conform to the dimensions and clearances specified in table II. The overall length and overall height of the semitrailer shall be the minimum practicable for the intended use, but shall not exceed the values listed in table II.

TABLE II. Dimensions and clearances.

Designation	Type I	Type II	Type III	Type IV
Overall length, feet (max.)	35	38	38	40
Overall width, inches (max.)	96	96	96	96
Overall height, inches (max.)	110	112	115	104
Ground clearance, inches (minimum)	11-1/2	11-1/2	11-1/2	11-1/2
Swing radius, from centerline of kingpin to most distant point on semitrailer nose, inches (max.)	60	60	60	60
Distance from centerline of kingpin to front end of semitrailer, inches (min.)	18	18	18	18
Turning clearance from centerline of kingpin to any portion of semitrailer 6 inches or more below upper fifth wheel plate, inches (min.)	82	82	82	82
Upper fifth wheel height, from ground to underside of plate, plus or minus one inch	49	49	49	49

3.2.2 Payloads and capacities. Payloads shall conform to table III. Hopper capacities, measured in accordance with SAE J741, shall conform to table III.

TABLE III. Payloads and capacities.

Designation	Size 18	Size 20	Size 22
Struck capacity, cubic yards (minimum)	18	20	22
Payload, pounds (minimum)	50,000	55,000	60,000

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3.3 Performance requirements.

3.3.1 Functional requirements. The semitrailer(s) shall be capable of transporting, dumping, spreading (types I and III) and windrowing (types II and IV) material, as specified in 3.5.1 through 3.5.2.

3.3.2 Mobility performance. The semitrailer, fully equipped and loaded with the specified payload (see 3.2.2), shall be capable of being towed at speeds as great as 10 miles per hour (mph) over unimproved roads and over reasonably hard uneven terrain; and being towed at speeds as great as 55 mph over improved roads.

3.3.2.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to the coupled towing vehicle without cramping or damaging the semitrailer or the towing vehicle.

3.3.2.2 Tracking ability. The semitrailer shall conform to the tracking requirements of Federal Motor Carrier Safety Regulation 393.70(a).

3.3.2.3 Brake performance. The service brakes shall stop the semitrailer-tractor combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Chassis frame and body structure. The semitrailer chassis frame and body structure shall be an electrically welded, all-steel integral unit. The rated payload and the sprung weight of the semitrailer shall not impose a fiber stress greater than 50 percent of the yield strength of the materials used when the semitrailer is operating under mobile operating conditions specified in 3.3. The chassis main frame members shall extend the full length of the body and shall be combined with sufficient number of crossmembers to insure stability.

3.4.2 Suspension system. The semitrailer shall be furnished with the manufacturer's standard suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, when the semitrailer is loaded with its rated payload. Clearances shall preclude interference between tires and any other part of the semitrailer under the operating conditions specified herein.

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3.4.3 Axles. The semitrailer shall be equipped with tandem axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the semitrailer loaded with the specified payload. The wheel bearings and axle spindles shall be oil lubricated. Oil viscosity shall be in accordance with the manufacturer's recommendations. The hub caps shall have a metal or heavy duty plastic body and a window for visual determination of oil level. Provision for venting, or equivalent method of withstanding pressures without leakage, and for replenishing the oil supply, shall be incorporated.

3.4.3.1 Undercarriage protection. Undercarriage protection shall be provided to prevent damage to axle, suspension and brake components from material during windrowing operations or spillage during loading.

3.4.4 Wheels, rims, tires and tubes. The semitrailer shall be equipped with dual wheels on each axle. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels of the semitrailer. When specified (see 6.2), disc type wheels shall be furnished.

3.4.4.1 Tires. Steel belted radial ply tires or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, when the semitrailer is loaded with the specified payload. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.4.2 Inner tubes. When tube type tires are furnished, inner tubes shall be heavy duty type, and shall be of proper size for the tires furnished.

3.4.4.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location under the chassis. Means shall be provided for securing the spare assembly within the carrier to prevent accidental loss.

3.4.4.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare tire assembly is specified (see 3.4.4.5), a spare wheel or rim shall be furnished.

3.4.4.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel

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or rim. The spare tire shall be of the same size, tread design and ply rating as the tires furnished on the vehicle.

3.4.5 Brakes.

3.4.5.1 Service brakes. Service brakes shall be of the full air, internal expanding type; shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45 through 393.47, and 393.52; and shall be as specified herein. The braking system shall include standard breakaway features, relay emergency valve, air reservoir, automatic slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains, and all other components required for a complete air-brake system. Gladhands shall conform to SAE J318. Location of air-hose fittings shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims. Slack adjusters shall be located above the bottom edge of the axle carrier.

3.4.5.2 Parking brakes. The semitrailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the semitrailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

3.4.6 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for coupling to a full oscillating and fore-and-aft rocking type fifth wheel, shall be of sufficient size to cover a fifth wheel 36 inches in diameter, and shall conform to Federal Motor Carrier Safety Regulation 393.70(b). The forward end of the upper fifth wheel plate shall have a turned-up lip for ease of coupling to a truck tractor and for body protection. The kingpin shall be fabricated of heat treated alloy steel and shall conform to SAE J700.

3.4.7 Landing supports. Landing supports, or stiff legs consisting of two legs of the gravity telescopic type mounted at the front of the semitrailer, shall be furnished. Each leg shall be equipped with a steel support pad not less than 12 inches square. Landing support adjustment shall be not less than 2-inch increments. The range of the adjustment shall vary the upper fifth wheel plate height from 47 inches to not less than 53 inches from the ground. Minimum ground clearance with support pads retracted shall be not less than 14 inches.

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3.4.8 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rail (unless recessed and fully protected) or on the semitrailer bumper. Turn signal lamps shall conform to SAE J588. All electrical wiring shall conform to SAE J1292. Clearance and identification lamps shall be constructed for easy removal and replacement of lamps and lenses without the use of hand tools.

3.4.8.1 Twelve-volt direct current (dc) system. A 12-volt dc lighting system conforming to 3.4.8 and 3.4.8.2 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.4.8.2 Receptacle for 12-volt dc lighting system. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.4.8.3 Independent 24-volt dc system. When specified (see 6.2), a 24-volt de lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.4.8. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.4.8.1, except the 24-volt dc stop and turn lamps shall be combined.

3.4.8.4 Interconnected 24-volt de system. When specified (see 6.2), capability of operating the 12-volt dc system (see 3.4.8.1) from a towing vehicle equipped with a nominal 24-volt de electrical system shall be provided.

3.4.8.5 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS75021, part number MS75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to left-hand turn signal and stop lamp (yellow)
Contact D. Connect to ground (white)
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown)

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Contact J. Connect to right-hand turn signal and stop lamp (green)
Contact L. Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

3.4.8.6 Resistors. When an interconnected 24-volt dc system is specified (see 3.4.8.4), the 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of the tactical towing vehicle from a nominal 24-volts dc to 12-volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to preclude overheating and any damage to resistors, wiring, or adjacent components.

3.4.9 Mud flaps. Manufacturer's standard mud flaps shall be provided to the rear of the rear wheels in accordance with SAE J682.

3.4.10 Rear end protection. Semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.4.10.1 Push plate. A rear push plate shall be provided on the rear of the semitrailer. The push plate shall withstand the forces encountered when pushing the fully loaded semitrailer, without breakage or permanent deformation of the plate or supporting members. The push plate may also provide rear under-ride protection.

3.5 Semitrailer types.

3.5.1 Type I and type II (single hopper, single gate). Type I and type II semitrailers shall have a hopper body with sides of sufficient vertical slope to insure complete dumping of contents through the air-operated gate located at the bottom of the hopper.

3.5.1.1 Hopper body. The hopper body shall be of all-welded fabrication, utilizing high-tensile steel having a minimum yield point of 70,000 pounds per square inch for all plates and

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reinforcing. Upper hopper interior sides and ends shall be smooth and shall slope into the lower dump hopper for smooth flow and complete cleanout of material.

3.5.2 Gate. An air-operated, spreader type (types I and III) or windrow type (types II and IV) clamshell gate shall be provided. The gate shall be mounted on fully lubricated, hinged bushings at the bottom of the lower dump hopper. The gate opening shall be large enough to allow complete unloading of the hopper body and shall be not less than 50 inches in width and not less than 90 inches in length for spreader type gate and not less than 52 inches in width and 96 inches in length for single gate windrow type. Adjustable spreader chains, adjustable stops, adjustment of throw of linkage arms by pins, or comparable adjustable controls shall be provided to permit spreading or sized windrowing.

3.5.2.1 Gate air system. The semitrailer shall be equipped with a complete air system capable of operating the gate(s) including cylinder(s), air reservoir(s), air line lubricator, and all necessary hoses and couplings, for connection to the truck-tractor air system. Air reservoir(s) shall be ASME coded for 125 pounds working pressure with not less than 6,900 cubic inch capacity for each gate. Cylinders, wall(s), and piston rod(s) shall be protected from corrosion.

3.5.2.2 Gate control. The semitrailer gate control shall be an electrical solenoid-air operated system. A receptacle suitable for electrical connection to the truck-tractor shall be installed on the front of the semitrailer. A switch kit shall be furnished for remote operation of the solenoid(s) from the truck-tractor cab. The switch kit shall include a plug compatible with the semitrailer receptacle, a connecting cable of sufficient length to extend from the semitrailer receptacle to the truck-tractor cab instrument panel, and a switch to be mounted on the instrument panel, accessible to the seated driver. A manual gate control air valve shall also be furnished on the left side of the hopper.

3.5.3 Type III and type IV (dual hopper, dual gate). Type III and type IV semitrailers shall have dual hopper bodies of equal size with sides of sufficient vertical slope to insure complete dumping of contents through an air-operated gate located at the bottom of each hopper. Construction of the hopper bodies shall be as specified in 3.5.1.1. Each gate shall be as specified in 3.5.2, except each windrow type gate shall be not less than 52 inches in width and not less than 62 inches in length. Gate air system shall be as specified in 3.5.2.1, but shall include separate

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cylinders, hoses, couplings and all other necessary provisions for independent operation of either gate, or simultaneous operation of both gates. Gate controls shall be as specified in 3.5.2.2, but shall include dual gate control switches and dual manual gate control air valve for independent operation of either gate.

3.6 Lubrication. Lubrication means shall be provided for all parts of the equipment normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals or other parts, fittings with pressure release shall be used.

3.7 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for immediate operational use. The servicing and adjusting shall include at least the following: inflation of all tires; adjustment of brakes; proper functioning of all lighting and controls; and complete lubrication, with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.8 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

3.8.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.8.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have all burrs removed. Appropriate washers or lockwashers shall be provided and all bolts, nuts and screws shall be tight or torqued in accordance with good commercial practice.

3.8.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.8.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the

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parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loadings. Fillet welds shall be provided when necessary to reduce stress concentration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. To determine conformance to all the requirements of the contract, the first production vehicle of each type (see 1.2) produced under the contract shall be subjected to inspection and a road test. The inspection and road test shall be conducted by the contractor, at or near his plant, under the direction and in the presence of Government representatives. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the contract.

4.3.1 Vehicle weight. The first production vehicle(s) shall be weighed to determine curb weight and distribution of curb weight on kingpin and rear axle(s). The imposed loading on the kingpin and rear axle(s) will be computed using the curb weight and the payload as specified in 3.2.2. Calculated imposed loads will be used to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contract requirements.

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4.3.2 Road test. To determine conformance to 3.3.2 through 3.3.2.3, the semitrailer, loaded with rated payload, and coupled to a truck tractor, shall be driven a distance of not less than 20 miles under the conditions specified in 3.3.2. At least 20 percent of the distance shall be over hard uneven terrain. The tractor semitrailer combination shall be subjected to not less than five sudden stops at a speed of not less than 20 mph and observed for the specified stopping distance. During turning ability operation, the tractor trailer combination shall be observed for cramping or damaging the semitrailer or towing vehicle. Tracking ability shall be observed for conformance to 3.3.2.3. Upon completion of the road test, the semitrailer shall be examined for leakage (air and lubricant), damage to air system, dumping system, evidence of misalinement, binding or other malfunction.

4.3.3 Landing support test. To determine conformance to 3.4.7, the landing supports shall be examined and tested. The landing supports shall be operated through the entire range of adjustments to assure that the upper fifth wheel plate height varies from 47 inches to not less than 53 inches from the ground. The support pads, in retracted position, shall be inspected for minimum ground clearance of not less than 14 inches.

4.3.4 Dump test. To determine conformance to 3.5.1 through 3.5.2, the semitrailer of each type shall be loaded and subjected to dumping and spreading or windrowing, as applicable. The controls operating the gate shall be used from the remote cab system and the manual controls at the left of the hopper(s). During the test the semitrailer shall be observed for proper operation of the air system and malfunction or binding of the dumping system. Upon completion of the test, the vehicle shall be inspected for permanent deformation, air leakage and lubrication leakage.

4.3.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.6 Failure. Failure of the first production vehicle to conform to any of the requirements specified in the contract shall be cause for rejection of the vehicle. Further, the Government may continue to refuse acceptance of all vehicles under the contract, until the contractor has taken corrective action to eliminate the cause(s) of the failure.

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4.4 Production vehicle inspection. Each production vehicle shall be subjected to dimensional inspection and performance tests (less payload) as specified in section 3. The contractor's inspection as a minimum, shall assure that the vehicle conforms to the dimensional and performance requirements, and the vehicle is capable of meeting the requirements herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government, indicating that the servicing and adjusting required by 3.7 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. Vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for use in transporting and bottom dumping materials such as clay, gravel and sand.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of vehicle required (see 1.2).
(c) Identification of the appropriate service for painting and marking (see 3.1.1.1).
(d) Rustproofing, if required (see 3.1.1.2).
(e) Lifting and tying down attachments, if required (see 3.1.1.3).
(f) Disc type wheels, if required (see 3.4.4).
(g) Bias ply tires, if required (see 3.4.4.1).
(h) Carrier for a spare tire assembly, if required (see 3.4.4.3).
(i) Spare wheel or rim, if required (see 3.4.4.4).
(j) Spare tire assembly, if required (see 3.4.4.5).
(k) Independent 24-volt dc system, if required (see 3.4.8.3).
(1) Interconnected 24-volt dc system, if required (see 3.4.8.4).

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2330-0817

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